The cos 2φ modulation on tensor-polarized ^{6}Li: projected measurements, the coherent intact-recoil channel, and what stands behind the numbers

2026-08-10 · development run 5 · polligen (evgen/) + polli_fastsim (fastsim/) · companion documents: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md (physics note) · all external numbers verified against primary sources in three research sweeps; adversarially reviewed

This report presents two projected measurements ("money plots") for a transversely tensor-polarized ^{6}Li beam at the EIC, explains the design choices behind the simulation machinery and the physics formulas, states every critical assumption explicitly, and lists the improvements that would firm the projections up. The observable is the cos 2φ azimuthal modulation driven by photon double-helicity flip — "exotic glue" [1,2] — measured two ways: inclusively (electron only), and in coherent diffraction with the ^{6}Li detected intact in its 1^+ ground state.

13.5% coherent-tag acceptance at IP6, high-acceptance optics ($\exp(-B\cdot p^2\_T{,}cut)$); 9–20% for $B \in$ 40–60 GeV^{-2}). High-divergence: 4×10^{-5} — the coherent program fixes the optics choice.	**1.1×10^7 / 10^8** tagged coherent events in the 1-year / 10-year programs (scenario f_0 = 0.04); best-bin $\delta A = 1.9\times10^{-3}$ / 6×10^{-4} → 5σ floors at 1.0% / 0.3% modulations	**0.036** deformation-anchored $\langle a_2\rangle$ of the tagged sample at P_zz = 0.6 [band 0.018–0.059], scaled from the polarized-deuteron CGC calculation [13] — sign flip vs the deuteron predicted

1 · The observable and why these formulas

For unpolarized electrons on a spin-1 target whose tensor polarization axis lies in the transverse plane, the inclusive cross section is [1,2]:

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dσ/(dx dy dϕ) = (2yα²/Q²) · [ F₁ + ⅓ a_m b₁ + (1−y)/(xy²) · (F₂ + ⅓ a_m b₂)
                         − (1−y)/y² · c_m sin²θ_m · Δ(x,Q²) · cos 2ϕ ]

a_m = ¼ c_m (3cos²θ_m − 1),   c_m = 3|λ_m| − 2 → (1, −2, 1)
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Why this observable. The cos 2φ term requires the virtual photon to flip helicity by two units. A spin-½ nucleon cannot absorb two units of helicity, so no quark — and no bound nucleon — contributes at leading twist: $\Delta(x,Q^2)$ is purely gluonic and vanishes for a nucleus that is "just protons and neutrons" [1]. It has never been measured for any target; the only prior experimental document is a JLab letter of intent with no sensitivity projection [3]. Magnitude guidance spans per-mille (Sather–Schmidt bag moment $-0.012\,\alpha_s$ [4]) to few-percent (maximal $\Delta_T g = \Delta g$, explicitly an overestimate [5]) — hence the scenario band $\Delta/F_1 \in \{10^{-3} \dots 10^{-2}\}$.

Formula choices made and their cost. The master formula is implemented in the Hoodbhoy–Jaffe–Manohar normalization [2] with the fastsim approximation set: $A_{/\!/} = D\cdot g_1/F_1$ (γ^2-suppressed η term dropped), $g_2 = g_2^{WW}$, and no target-mass/γ^2 corrections. The last is a real limitation only at high x and low Q^2; the sweet-spot bins sit at $x \le 0.06$ where $\gamma^2 = 4M^2x^2/Q^2 < 10^{-2}$ even at $Q^2 \approx 1$ GeV^2, so it is a labeling caveat, not a bias, for these plots. Structure functions are TOY parameterizations with PDF-grid (CT18NLO/NNPDFpol11) backends behind the same interface — rates are trustworthy to a factor ~1.5, asymmetry *shapes* are scenarios by construction.

2 · Simulation design choices

Why an event-level machinery at all. No public generator produces polarized e+A events for A > 1 (verified survey; BeAGLE is unpolarized, DJANGOH/PEPSI are nucleon-level longitudinal-only). polligen implements only the missing wheel — the spin-density matrix ⊗ structure-function master formula and spin-correlated sampling — and reuses everything else. Its Step-5.A validation gates: bin-wise identity with the analytic asymmetry formulas at rtol 10^{-12}, pseudo-experiment error spreads closing on every analytic FOM map, and unbiased cos 2φ recovery with holey φ acceptance.

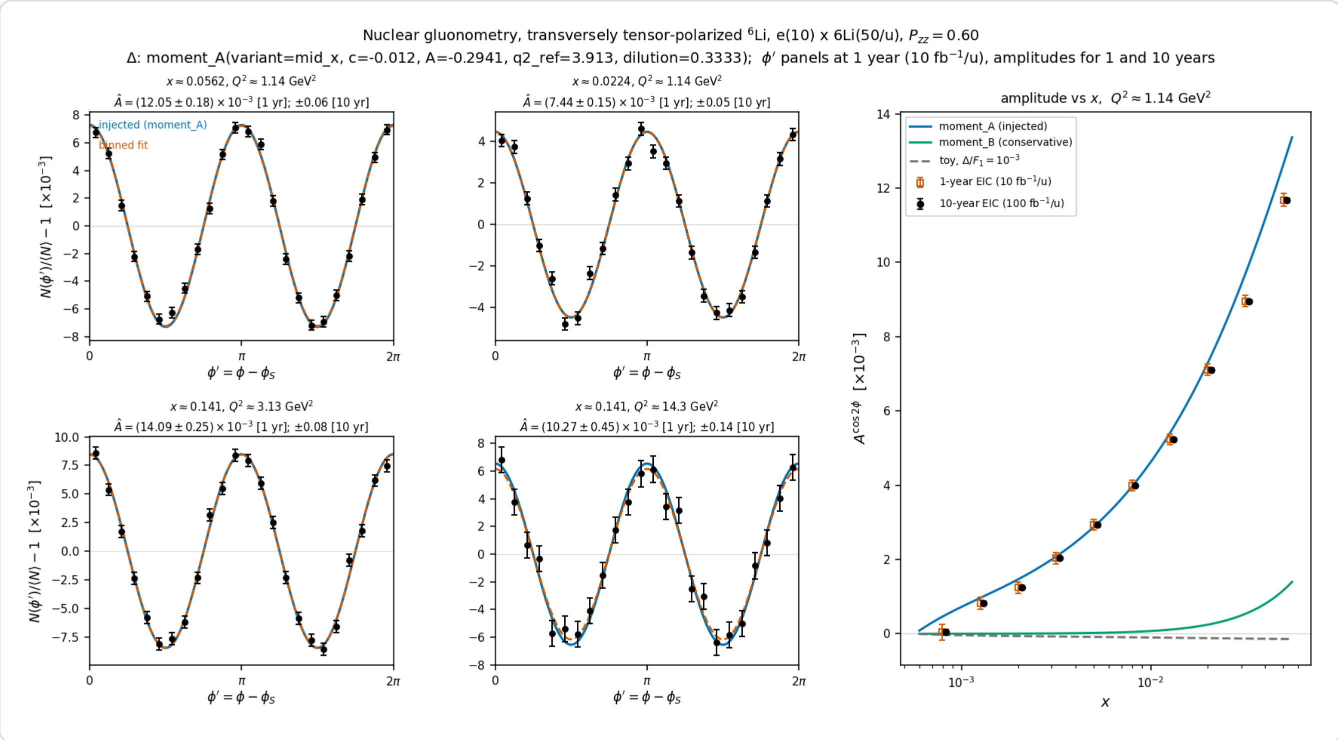
Full-luminosity statistics without 10^9-event files. The money plots need pseudo-data at N up to ~10^9 events per bin. For a *binned* estimator, drawing each φ-bin's count from a Poisson with the exact expected yield (the analytic integral of the modulated cross section over the bin) is statistically identical to event-level sampling — so a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`). Event-level sampling remains available and is what the closure tests use.

Estimator choice. The amplitude is extracted by a binned least-squares fit of $N(\varphi') = C(1 + A \cos 2\varphi')$ (`estimators.cos2phi_fit_binned`), not the naive $2\langle\cos 2\varphi'\rangle$ moment: with realistic φ-acceptance holes the moment is demonstrably biased by more than the signal, while the fit — with bins probed across their full width and dropped if they straddle a hole — stays unbiased. Statistical errors follow $\delta A = \sqrt{(2/N)}/P_{zz}$, validated by pseudo-experiment spreads. A precondition is documented: the fit model carries no cos φ′ term; the m-symmetric tensor fills guarantee $a_1 = 0$ identically by design.

Sweet-spot selection. The four displayed (x, Q^2) super-bins are chosen from the analytic significance map *per Q^2 band* rather than by raw ranking — the raw map clusters at the lowest Q^2 (rate-dominated), but the Q^2 lever arm is itself physics: it is the consistency check against power-suppressed non-partonic contributions (§5). Each super-bin merges 3×3 cells of the analysis grid (~×2 in x and Q^2).

Rate conventions. Per-nucleon luminosity and F_2A/A structure functions throughout; scattered-electron acceptance cuts ($|\eta| \le 3.5$, $E' \ge 0.5$ GeV, $0.01 \le y \le 0.95$, $W^2 \ge 10$ GeV^2) applied in the rate maps; the whole luminosity is assigned to the transverse-tensor fill — the same convention as the analytic FOM maps (a run-plan split would scale N down proportionally). Projections are quoted separately for the **1-year (10 fb^{-1}/u) and 10-year (100 fb^{-1}/u) EIC programs**.

One home for the Δ model. All $\Delta(x,Q^2)$ models live in a unified registry, `polli_fastsim/delta_models.py` , behind the single (x, Q^2, F_1) interface every kernel consumes — new constraints and models land in one file and every consumer switches by name. The production default is `moment_A` , the sum-rule-constrained ansatz of the merged money_delta study line: $\Delta = A\cdot\alpha_s(Q^2)\cdot F_1\cdot x^a(1-x)^b$ with A solved so that $\int x\,\Delta\,dx = -0.012\cdot\alpha_s$ (the Sather–Schmidt bag moment [4]; solved $A = -0.294$ at $\langle Q^2\rangle = 3.9$ GeV^2, matching the money_delta EPPS21 value −0.31), times the ^{6}Li per-nucleon dilution 1/3 — the Cloët convention, kept separate from the whole-nucleus P_{zz} so the two never double-count. `moment_B` (no F_1 factor; the conservative reading, a factor ~2–8 below A) and the legacy `toy` shape are one flag away; the A-vs-B spread is the dominant ansatz systematic, and the amplitude-vs-x panel shows both — under the moment constraint the measurement *resolves the interpretation*, not just a null.



Money plot 5 — inclusive gluonometry (e10 × ^{6}Li50, P_zz = 0.60, Δ model: moment_A, dilution 1/3). Left 2×2: φ′-modulation pseudo-data with statistical error bars at the **1-year program** (10 fb^{-1}/u) in the four sweet-spot super-bins (Q^2 = 1.1 → 14 GeV^2, x ≈ 0.02–0.14); injected curve (blue), binned fit (vermillion); per-panel annotations give δA for both programs. Right: extracted amplitude vs x with 1-year (open) and 10-year (filled) error bars against the moment_A (blue), moment_B (green), and toy-10^{-3} (gray) curves. Amplitudes are (0.7–1.4)×10^{-2} at the sweet spots against δA = (1.5–4.5)×10^{-4} in year 1 — every bin > 25σ, so the A-vs-B ansatz spread is resolved by the x-shape. Detection needs only the scattered electron.

3 · The coherent channel: model choices and the anchored amplitude

Tagging. The intact ^{6}Li recoil has A/Z = 2 — exactly beam rigidity (R = 1.000; the diffractive momentum loss x_P < 0.01 changes rigidity by < 1%). At IP6 it is visible *only* in the Roman-Pot near-beam p_T tail above the 10σ beam-envelope cut [7,9]. With an exponential coherent t-distribution the acceptance is analytic, exp(−B·p²_T,cut): **13.5%** for the documented high-acceptance cut (p_T > 0.20 GeV), **4×10⁻⁵** for high-divergence (0.45 GeV — a derived, not documented, number) — the coherent program fixes the optics choice. The IR-8 secondary focus reaches p_T ≈ 0; published intact-recoil efficiencies (d 47%, ^{3}He 32%, ^{4}He 29%, ^{7}Li 17.8% [10]) interpolate to ≈20% for ^{6}Li — an independent cross-check.

Rate model choices (`polligen/coherent.py`, all bands explicit): σ_coh = f_coh(x)·σ_DIS with f_coh = f₀/(1+(x/x_coh)²), f₀ = 0.04 [0.02–0.08] — no light-nucleus prediction exists (lightest published: Ca), so the band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-A saturation estimates (20–25% [45]); x_coh = 0.01 from the coherence length l_c = 0.105 fm/x_P vs the ^{6}Li size; slope B = 50 GeV⁻² [40–60] from B = ⟨r²⟩/3 with the matter radius (charge radius gives ≈57 [47]; the gluonic B may sit higher [42]), exponential valid below the form-factor minimum at |t| ≈ 0.31 GeV² [11,12]. Writing f_coh against Bjorken x (not x_P) is optimistic for high-mass diffraction; the f₀ band covers it. Tagged yields: 1.1×10⁷ events in year 1 (10 fb⁻¹/u), 1.1×10⁸ over ten years; best super-bin δA = 1.9×10⁻³ / 6×10⁻⁴ (1 yr / 10 yr).

The amplitude is anchored, not guessed. The polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13] is the *only* published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all 8 forward citations checked; the group's follow-ups are unpolarized [14,15]). Their figures, digitized from the arXiv source into `coherent.MANTYSAARI_A2_DEUTERON`: a2(m=0) = +0.08/+0.15/+0.30/+0.43 at |t| = 0.05/0.1/0.2/0.3 GeV², a2(m=±1) ≈ −a2(m=0)/2, a factor-2 m-state cross-section difference at |t| = 0.1 — all from the 6% D-wave modulating the diffractive slope, a2 ≈ (ΔB_m/2)·|t| with ΔB₀/B ≈ 0.21. For a fill with populations p_m the ensemble coefficient is exactly ∝ P_zz:

```
a₂(t; P_zz) = -(P_zz/4) · ε_B0 · B · |t|,      ε_B0 ≡ ΔB₀/B

⁶Li scaling of ε_B0 (two dials that usefully disagree):
  · relative quadrupole deformation Q/⟨r²⟩:  ⁶Li ≈ (1/5) × deuteron, OPPOSITE sign
  · α-d asymptotic D/S ratio η = -0.010…-0.015 [16]  vs  η_d = +0.0256:  0.4-0.6×
  ⇒ ε_B0(⁶Li) ∈ -(0.04 … 0.13),  default -0.08
  ⇒ ⟨a₂⟩_tag = a₂(⟨|t|⟩_tag = p²_T,cut + 1/B) ≈ 0.036 [0.018-0.059] at P_zz = 0.6
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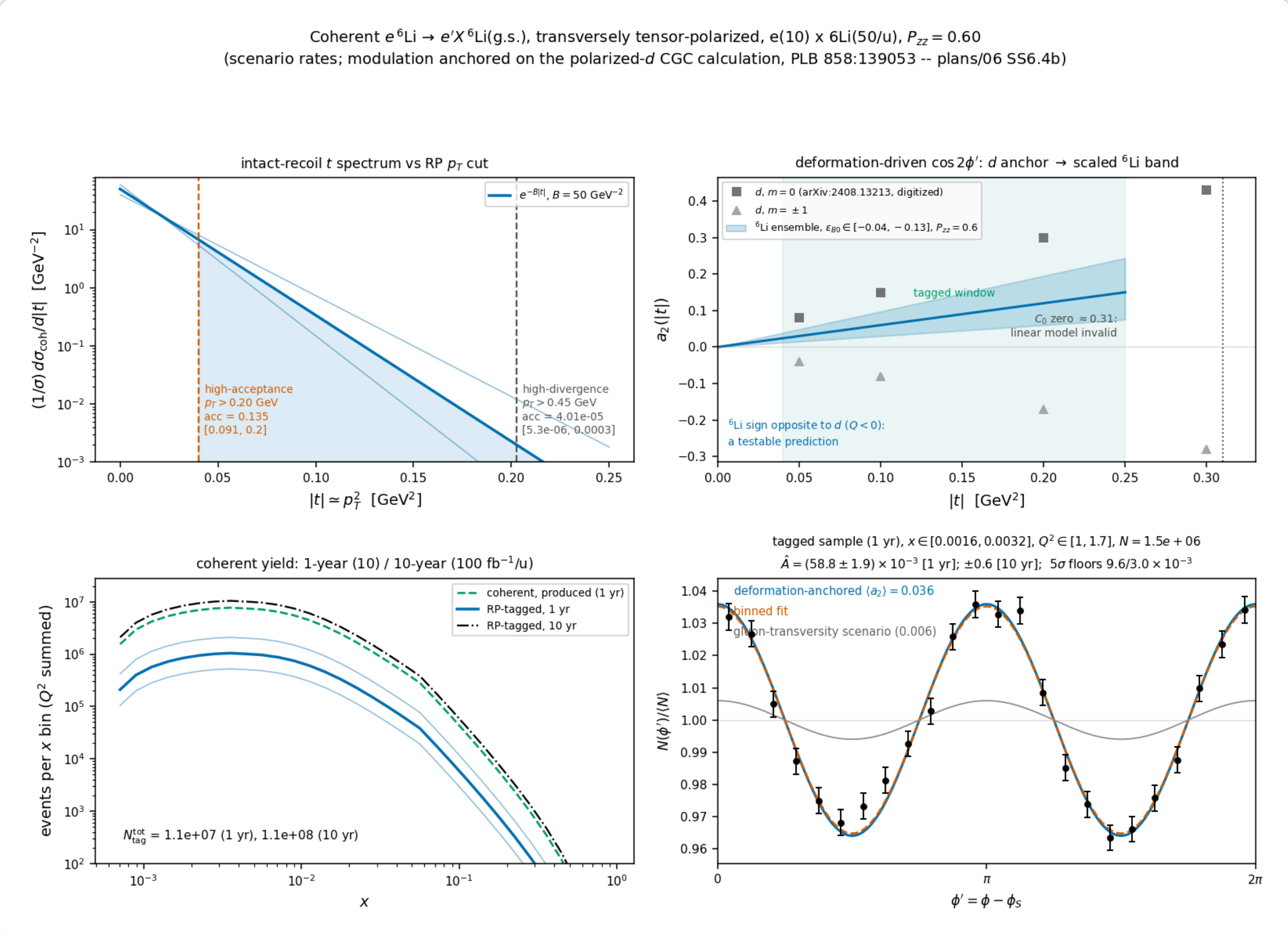
The **sign flip relative to the deuteron is a testable prediction**. The band is physics, not sloppiness: even exact few-body theory misses the ^{6}Li quadrupole cancellation by ~3× [17], and the gluonic deformation need not cancel like the charge one. Existence proofs at exactly this |t|: measured e-d $T_{20} = -(0.1\text{–}0.3)$ [19]; HERMES A_zz(d) ≈ −1% at x ≈ 0.01 [20], coherent-double-scattering in origin [21–23]. The flat gluon-transversity term is separately bounded at 3×10^{-3}–10^{-2} (lattice φ-meson transversity [25,26] ÷ the ~10× NPLQCD nuclear suppression [27]; Kumano–Song few-% only at maximal Δ_T g [5,28]).

> **The measurement this buys:** the two mechanisms separate by shape — deformation is t-linear, sign-flipped vs the deuteron, and should weaken toward small x_P (JIMWLK washes out deformation [15]); exotic glue is flat in t and survives at small x. A two-component fit $a_2(t) = c_{def}\cdot|t| + a_g$ over the tagged window, per x_P bin, with both components ≥ 5σ above the statistical floor at 100 fb⁻¹/u. Incoherent breakup is m-state-blind in this mechanism [13] — it dilutes but cannot fake the modulation.



Money plot 6 — coherent channel, e ^{6}Li → e′ X ^{6}Li(g.s.) (same beam, P_zz = 0.60). Top left: recoil t-spectrum vs both optics' p_T cuts with the slope band. Top right: the amplitude anchor — digitized deuteron a_2 per m-state [13] vs the scaled ^{6}Li ensemble band (sign flip predicted); tagged window and C_0-zero exclusion marked. Bottom left: tagged yield vs x for the 1-year and 10-year programs with the f_0 band (N_tag ≈ 1.1×10^7 / 1.1×10^8). Bottom right: φ′ pseudo-data (1-year statistics) at the deformation-anchored ⟨a_2⟩ = 0.036 with the flat gluon-transversity scenario (gray); extraction Â ± 1.9×10^{-3} (1 yr), ± 0.6×10^{-3} (10 yr).

4 · Backgrounds for the coherent tag (summary — full budget in plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
α + d breakup (threshold 1.474 MeV; 2.186 MeV 3^+, Γ = 24 keV, ~100% α+d [39])	Both fragments at R = `1.000`, same trajectory *and* velocity as intact ^{6}Li — tracking and ToF blind; only dE/dx (∝ Z^2: 9/4/1) separates	RP charge discrimination — open question #19: no EIC document addresses the A/Z = 2 degeneracy (AC-LGAD amplitude exists but is used for position; documented Z-ID is an IR-8 Cherenkov [8]). Fallback: two-component |t| fit (coherent $e^{-50\|t\|}$ vs ~10× flatter incoherent; crossover at 0.05–0.2 GeV² sits inside the window [41,42]); e+Pb benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: α+p+n (3.70 MeV), ^{3}He+t (15.79 MeV) [39]	A fragment could pass unnoticed...	...but p → OMD (R = 0.50), n → ZDC, ^{3}He → RP main window (R = 0.75): vetoable; inefficiency is second-order (BeAGLE study to quantify)
^{6}Li* 3.5629 MeV (0^+, T = 1, Γ = 8.2 eV) [39]	Particle-stable (α+d strictly parity-forbidden, Γ_α ≤ 6×10^{-7} eV); γ-decays → real A=6, Z=3 track	Isoscalar exchange cannot drive T=0 → T=1; the boosted 0.1–0.4 GeV γ lands mostly in the B0 EMCal (≥ 50–100 MeV, 5.5–20 mrad [9]), tail in ZDC. The 5.366 MeV 2^+ T=1 always emits a nucleon → vetoable. **Every T=1 state is γ- or nucleon-vetoable; only T=0 states feed the blind α+d channel**
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t-dependence = charge FF² [11,12]); subtract + reuse as RP-acceptance candle; charge-quadrupole cos 2φ < 10^{-3} for |t| < 0.15 GeV² (^{6}Li C2 unobservably small below q ≈ 3 fm^{-1} [12,17])
Beam-gas / halo; e′-side fakes	Accidental R ≈ 1 tracks; π/pair-symmetric e′ at high y	Non-colliding bunches, ~30 ps timing, vertex association; standard DIS e′ ID + (e′, recoil) kinematic consistency

Extraction systematics: the modulation is self-normalized within a single fill — relative-luminosity systematics drop out; φ-acceptance holes are handled by the binned fit (demonstrated); polarimetry scale δP_zz/P_zz ≈ 3% is common with A_zz; tensor-observable radiative corrections are uncharted and flagged on every plot. Interpretation-level: a ΔΔ-isobar admixture can mimic Δ at leading twist [34]; conventional double-scattering generates the low-x b_1 [21–23] — the Q² lever arm and the ^{6}Li/d comparison are the discriminants.

> **The ^{6}Li null test:** deformation-driven coherent modulations — the polarized-deuteron imaging signal [13] — are bounded on ^{6}Li by its anomalously small quadrupole moment, Q(^{6}Li) = −0.0806(6) fm² ≈ Q(^{7}Li)/50 [35]. A sizable flat-in-t cos 2φ on ^{6}Li cannot be a shape effect: the mechanisms entangled for the deuteron separate cleanly in the ^{6}Li/d comparison.

5 · Critical assumptions, honestly ranked

#	ASSUMPTION	IMPACT IF WRONG	REPLACEMENT PATH
1	Scenario coherent fraction f_0 = 0.04 [×2/÷2] and its Bjorken-x (not x_P) parameterization	All coherent yields scale linearly; δA scales as $1/\sqrt{f_0}$	Real diffractive model — open question #18; the concrete ask is an IP-Glasma α–d run [13]
2	ε_B0 scaling from the deuteron (two dials disagreeing by ×3; charge-quadrupole cancellation may not transfer to gluons)	Deformation amplitude anywhere in the stated band; sign flip robust	Same IP-Glasma α–d calculation; meanwhile the band is quoted on every plot
3	Event-by-event Z-ID in the RPs is NOT assumed (purity via |t|-shape fit)	If Z-ID exists, purity improves outright; if the incoherent slope model is off, the fit systematic grows	#19 answer from ePIC FF WG; BeAGLE Mode-W incoherent spectra
4	Linear $a_2(t)$ valid only for |t| < 0.2 GeV²; C_0 zero at 0.31 GeV² excluded, not modeled	Upper tagged window carries unmodeled structure (locally enhanced, sign-flipping a_2)	Form-factor-level model of the dip region (the deuteron analog is in [13])
5	TOY structure functions; no target-mass terms; $g_2 = g_2$WW	Rates to ×1.5; shapes are scenarios by construction	PDF-grid backends exist; digitized theory curves scheduled (plans/02 step 1.2)
6	High-divergence p_T cut 0.45 GeV is derived from YR divergence tables, not documented	Only affects the (already excluded) high-divergence option	ePIC optics documents as they firm up
7	All luminosity in the transverse-tensor fill; P_zz = 0.6 in-ring placeholder	N scales with the run-plan share; amplitude scales with P_zz	Run-plan bookkeeper already supports splits; EPIOS/source numbers as they land (ABS

			sources: ~95% of maximal P_zz demonstrated [37])
8	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger |t|; RC unquantified for tensor observables	Theory engagement (plans/04 #10, #16); QED tail is calculable and doubles as calibration
9	Deformation term assumed x_P-independent within the plotted bins	JIMWLK evolution weakens deformation toward small x [15] — would mimic a relative growth of the gluonic term	Make the x_P dependence part of the two-component fit; theory input from [15]

6 · Future improvements

Near term (this program): BeAGLE Mode-W incoherent ^{6}Li spectra → veto-efficiency and |t|-fit systematics become numbers (plans/02 step 1.5); VMC α–d/α–t overlaps replace the two-parameter radial forms (plans/04 #15); pseudo-experiment closure with a 2-D (φ, φ_t) acceptance map (Phase-2, plans/03 step 2.3); HepMC3 output + ePIC chain smoke test (plans/05 step 5.D) so these projections can be repeated at reconstructed level.

Theory engagement: #18 — rerun the IP-Glasma polarized-deuteron setup [13] with an α–d cluster density for ^{6}Li (the single highest-leverage input: it would replace both ε_B0 dials and the f₀ band at once); the spin-3/2 (^{7}Li) structure-function basis (plans/04 #14); tensor-observable radiative corrections (#10); the sum-rule tie between the transversity-GPD moments and the measured tensor form factors [30] to implement the cancellation scenario rigorously.

Detector/machine questions raised by this study: #19 — can the RP AC-LGAD amplitude separate Z = 3/2/1 at the same rigidity and velocity? (No EIC document addresses it; decides whether coherent-tag purity is event-by-event or statistical.) The IR-8 secondary focus with its Z^2 Cherenkov [8] is the structural fix and lifts the tag acceptance from 13.5% toward pT ≈ 0 coverage (^{7}Li: 17.8% [10]) — the coherent-^{6}Li program is one more physics case for the second IR. Off-momentum halo determines how close the pots sit, i.e. the pT cut itself.

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Generated from the PolarizedLithiumSim repository (development run 5, 2026-08-10): `evgen/scripts/money_cos2phi.py`, `evgen/scripts/money_cos2phi_coherent.py`, models in `polli_fastsim/delta_models.py` (unified Δ registry) and `evgen/polligen/coherent.py`; 66 + 22 tests pass, including the merged money_delta study line. All rates use TOY/scenario inputs — bands, not predictions. External numbers verified against primary sources in three research sweeps (nuclear data: TUNL; detector: EIC YR + ePIC documents; theory: original papers); citation metadata individually verified against arXiv/INSPIRE. Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.